Table 1. Recall@100 over integrated vector databases

$\texttt{emb}_1$ **vs** $\texttt{emb}_2$	**method**	**Dataset**			
		SciFact	NFCorpus	ArguAna	SciDocs
Fast Text **vs** NV-embed V2	Union	26.33	13.62	30.34	10.80
	A2M	18.33	8.05	25.91	5.70
	LA2M	76.67	34.06	64.60	39.30
NV-embed **vs** Openai Ada	Union	32.00	14.24	28.19	16.70
	A2M	32.00	14.24	28.19	16.70
	LA2M	79.33	35.29	85.22	41.10
gte-Qwen2 **vs** NV-embed V2	Union	31.67	18.27	28.48	19.60
	A2M	32.33	15.48	28.48	18.40
	LA2M	85.33	31.27	86.65	44.60
Mistral **vs** Openai Ada	Union	32.00	15.17	28.19	17.60
	A2M	27.00	10.84	29.91	12.60
	LA2M	76.67	34.06	79.23	42.00

4 Hypothesis Revisited for Vector Database Integration

From Section 3, we observe that Hypothesis A does not necessarily hold for vector databases. As a result, one cannot simply reduce the integration of vector databases to an application of existing embedding alignment techniques via `A2M` as this would yield an integrated database that cannot accurately implement top-k similarity search.

To rectify this, we first re-examine the hypothesis, showing that all we need to make Hypothesis A work for vector databases is a simple condition on vector distances of concern (Section 4.1). Based on it we then develop a new integration framework (Section 4.2).

4.1 Hypothesis: From Global Isometry to Local Isometries

We have seen from Section 3.2 that the integration of vector databases $\mathcal{D}_1$ and $\mathcal{D}_2$ for data items of $\mathcal{O}_1$ and $\mathcal{O}_2$, respectively, cannot serve as a vector database that encodes the universe $\mathcal{O}_1 \cup \mathcal{O}_2$, as shown by the performance of `A2M` in Table 1. This is mostly due to that different embedding models produce non-isometric vectors for common data items in $\mathcal{O}_1 \cap \mathcal{O}_2$ as shown in Fig. 3a of Section 3.1.

However, we are not completely fruitless. In fact, the experimental findings, especially those in Fig. 3b, seem to suggest that different embedding models behave quite consistently for *short distances*. Such short distances are also the key to implement accurate top-k similarity search for reasonable k.

As shown in Fig. 3b, over SciFact dataset when k is small, *e.g.*, less than 5, top-k document retrieval results with `NV-Embed-V2` embeddings and those with `OpenAI-Ada` are almost identical, with Jaccard Index consistently approaching 1. When k increases, this consistency between `NV-Embed-V2` and `OpenAI-Ada`decreases significantly. This seems to suggest that `NV-Embed-V2` and `OpenAI-Ada` encode closely related documents via consistent short vector distances, despite that they are entirely different embedding models. We capture this observation via a new hypothesis below.

Hypothesis B [The Local Isometry Hypothesis]: *The global isometry hypothesis (Hypothesis A) holds within local regions of vector databases comprised of vectors with short distances.* □

To validate if this holds, we first visually examine local structures of vector databases embedded by `FastText` [8] and `NV-Embed-V2` [50] for the SciFact dataset. The zoomed-in T-SNE visualization of a query and its 10 nearest neighbors is shown in Fig. 4, where the left side depicts the distribution of results identified by top-10 vector search over `FastText` embedding vectors, and the right side shows the results of `NV-Embed-V2` vectors. It is not difficult to see that the distribution of the top-k vector search results of the two embedding models are strikingly similarly: they produce exactly

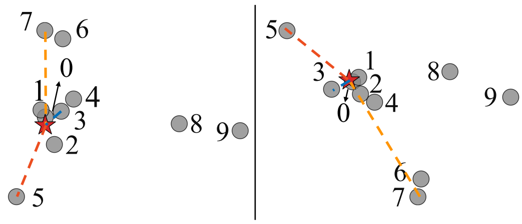

Fig. 4. A zoomed-in T-SNE view of a query (★) and its 10 nearest neighbors in SciFact **(**left shows the results over `FastText` [91] embeddings and right depicts `NV-Embed-V2` [50] results.)

the same list of top-10 most similar SciFact documents, and these answers form an approximately isometric shape in the vector space. This seems to suggest that Hypothesis A(1) indeed holds in this local region of SciFact vectors embedded by `FastText` and `NV-Embed-V2`

To further evaluate if Hypothesis A(2) holds in local regions as Hypothesis B states above, we note that Hypothesis B naturally suggests a new approach to integrating vector databases produced by different embedding models, which we refer to as *localized* `A2M` (denoted by `LA2M` and illustrated in Fig. 2c): (1) we first decompose vector databases into clusters of vectors that have short inter-distances; (2) we then apply `A2M` of Section 3.2 independently to each cluster.

We then test the accuracy of integrated vector databases computed by `LA2M`, by using exactly the same setting as we evaluate `A2M` in Section 3.2. The results, as highlighted in gray in Table 1, show that `LA2M` significantly boosts the accuracy of `A2M`. For instance, its recall@100 of document retrieval results over integrated vector databases is consistently more than double of that of `A2M` and `Union` (see more results in Section 7). This further confirms that Hypothesis A(2) indeed holds in local regions of vector databases. These findings indicate that, for real-life vector databases, Hypothesis B is significantly more valid than Hypothesis A.

4.2 A Framework for Vector Database Integration

We next describe framework `LA2M` in detail, showing how we make "localized" use of embedding alignment to best exploit Hypothesis B for better integrated databases. We also point out key technical challenges in using `LA2M`, which we will tackle in subsequent sections.

Framework. `LA2M` takes as input two vector databases $\mathcal{D}_1$ and $\mathcal{D}_2$ and produces an integration mapping function $\mathcal{T}$ such that $\mathcal{T}(\mathcal{D}_1) \cup \mathcal{D}_2$ serves as an integrated vector database for data items encoded by $\mathcal{D}_1$ and $\mathcal{D}_2$ taken together. Here $\mathcal{D}_1$ and $\mathcal{D}_2$ are possibly generated by two different embedding models $\texttt{emb}_1$ and $\texttt{emb}_2$, respectively, for data item collections O_1 and O_2; however, `LA2M` assumes no access or knowledge to either $\texttt{emb}_i$ or $O_i (i \in \{1, 2\})$ when computing the integration function $\mathcal{T}$.

It does so by exploiting *reference* vectors between $\mathcal{D}_1$ and $\mathcal{D}_2$: pairs of vectors that encode the same objects, *i.e.,* $\mathcal{X}$ and $\mathcal{Y}$ in `A2M` of Section 3.2. Reference vectors are abundant in scenarios where we need to integrate vector databases, as databases must be relevant so that integration is beneficial. Typical sources of references include known correspondence in embedding backfilling, domain knowledge, or data linkage results. In case they are not know as a priori, `LA2M` allows to automatically deduce such references (Section 6).

More specifically, `LA2M` integrates $\mathcal{D}_1$ into $\mathcal{D}_2$ in three steps:

Step (1): *Grouping*. It first groups the reference vectors $\mathcal{X}$ in $\mathcal{D}_1$ into clusters $X_1, \ldots, X_n$, each with vectors that have short inter-distances. Denote by Y_i the set of matching vectors of X_i in $\mathcal{Y}$ of $\mathcal{D}_2$ ($i \in [1, n]$). Note that the clusters do not have to be disjoint.

Step (2): *Alignment*. Following Hypothesis B, LA2M deduces a collection of isometries to align the reference clusters independently. For each pair X_i and Y_i, it invokes A2M to compute an isometry f^i_{iso} that aligns X_i to Y_i as discussed in Section 3.2. One may also replace A2M with other embedding alignment variants; we use isometries here as they are easy to compute and offer us a provable bound on the quality of integrated vector database as we will see in Section 5.

Step (3): *Merging*. LA2M decides the integration mapping $\mathcal{T}$ for $\mathcal{D}_1$ with local isometries computed in step (2). Specifically, for any vector $\mathbf{x}$ in $\mathcal{D}_1$, $\mathcal{T}(\mathbf{x})$ maps $\mathbf{x}$ according to the position of $\mathbf{x}$:

(a) $\mathbf{x}$ *is a reference vector*. When $\mathbf{x}$ is a reference vector in a cluster X_i grouped in step (1), $\mathcal{T}$ maps $\mathbf{x}$ to the vector $\mathbf{y}$ in Y_i of $\mathcal{D}_2$ that encodes the same data item as $\mathbf{x}$ does in X_i of $\mathcal{D}_1$.

(b) $\mathbf{x}$ *is a non-reference vector*. Otherwise, $\mathbf{x}$ is a non-reference vector in $\mathcal{D}_1$ that needs to be mapped to the vector space of $\mathcal{D}_2$. $\mathcal{T}$ does so by picking X_{j^*} with centroid $\mathbf{c}_{j^*}$ that is closest to $\mathbf{x}$ among all reference clusters, *i.e.,* $j^* = \arg\min_{i\in[1,n]} \|\mathbf{c}_i - \mathbf{x}\|$, where $\mathbf{c}_{j^*} = \sum_{\mathbf{x}\in X_{j^*}} \mathbf{x}/|X_{j^*}|$. It then uses $f^{j^*}_{\text{iso}}$ to map $\mathbf{x}$ to $\mathcal{D}_2$, *i.e.,* $\mathcal{T}(\mathbf{x}) = f^{j}_{\text{iso}}(\mathbf{x})$.

Finally, LA2M returns $\mathcal{T}(\mathcal{D}_1) \cup (\mathcal{D}_2 \setminus \mathcal{Y})$ as the integrated database. With access to $\mathcal{D}_1$ and $\mathcal{D}_2$ only, LA2M expands the scope of top-k similarity search from over only data items encoded by $\mathcal{D}_2$ to over the universe of objects encoded by $\mathcal{D}_1$ and $\mathcal{D}_2$ taken together.

Benefits & implications. As shown in Table 1, LA2M enables us to integrate vector databases across embedding models, realizing cross-dataset similarity search. It also has diverse implications.

(1) Model & object oblivious. LA2M merges vector databases without integrating the data items encoded by the vectors, nor requiring access to the embedding models. Such obliviousness to the actual encoded datasets and embedding models makes LA2M a perfect fit for cases where we cannot freely share objects but still want to search across datasets, *e.g.,* when sharing data objects can be expensive such as in edge devices or restricted due to privacy constraints.

(2) Vector index invariant. LA2M not only transforms and maps vectors of $\mathcal{D}_1$ to $\mathcal{D}_2$, it can also directly transforms vector indices built atop $\mathcal{D}_1$, *e.g.,* IVF and HNSW, to the transformed vectors in $\mathcal{D}_2$ space. This is because LA2M uses isometries to align and merge vectors. In addition to preserving similarity search over $\mathcal{D}_1$, isometries can also transform IVF and HNSW over $\mathcal{D}_1$ as they are based on vector distances which are preserved during the merging step.

Key factors & challenges. The quality of the integrated database $\mathcal{T}(\mathcal{D}_1) \cup \mathcal{D}_2$ heavily relies on the following two factors of LA2M:

- *Availability of references* $\mathcal{X} \subseteq \mathcal{D}_1$ and $\mathcal{Y} \subseteq \mathcal{D}_2$.
- *Scope of isometries*: grouping $\mathcal{X}$ in to $X_1, \ldots, X_n$.

LA2M decides the isometries by analyzing reference vectors $\mathcal{X}$ and $\mathcal{Y}$ across $\mathcal{D}_1$ and $\mathcal{D}_2$. Such references are available from domain knowledge on the datasets, or can be deduced by employing entity linking methods [10, 71]. The rationale is that when we want to integrate $\mathcal{D}_1$ and $\mathcal{D}_2$, the data items encoded by them must be correlated; such correlation offers potential for the existence of linkage across the datasets, which, mapped to the vectors, gives us the reference vectors $\mathcal{X}$ and $\mathcal{Y}$. In the worst case where $\mathcal{X}$ and $\mathcal{Y}$ are not available, we develop an interface for LA2M to actively generate references with minimum access to the datasets (Section 6.2).

Even more challenging is to decide how we shall group $\mathcal{X}$ into local clusters $X_1, \ldots, X_n$, to maximize the quality of integrated vector database $\mathcal{T}(\mathcal{D}_1) \cup \mathcal{D}_2$. As shown in Fig. 5, the scope of X_i evidently affects the errors that LA2M produces in step (2) and step (3), which we refer to as the *alignment* error α and *integration* error β, respectively. To deal with this, we first theoretically

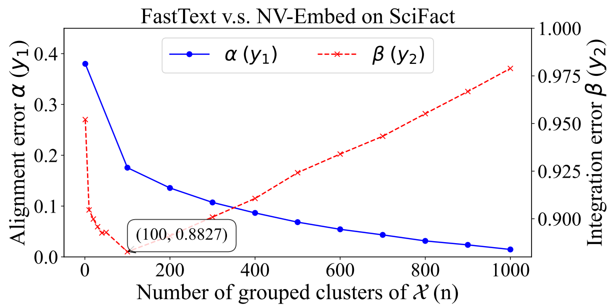

Fig. 5. #-clusters of $\mathcal{X}$ vs alignment error α and integration error β. (alignment error α: distance between $f^j_{\text{iso}}(\mathbf{x})$ and y when $\mathbf{x} \in X_j$ corresponds to $\mathbf{y} \in Y_j$ (step (2) error); β: distance between $\mathcal{T}(\mathbf{x})$ and the vector would have been embedded by $\texttt{emb}_2$ (step (3) error). $\mathcal{X}$ is grouped with k-meaning clustering.)

analyze and bound the errors on mappings $\mathcal{T}(\mathbf{x})$ of vectors $\mathbf{x} \in \mathcal{D}_1$ in Section 5, based on which we then discuss how LA2M breaks down $\mathcal{X}$ in Section 6.1.

5 Bounding the Quality of Integrated Vector Databases

To decide key design factors of LA2M, we study the quality of integrated vector databases.

Quantifying integration quality. We first consider metrics to quantify the quality of vector databases integrated by LA2M. While recall@k as reported in Table 1 (and also Section 7) is a standard choice for similarity search, it is sensitive to the curated query-answer samples built in the benchmarks; moreover, its empirical nature offers little insight in the source and properties of the errors.

To this end, below we use vector distances directly to quantify the error of every transformed vector of $\mathcal{D}_1$ in the vector space of $\mathcal{D}_2$. Recall from Section 4.2 that LA2M transforms vectors of $\mathcal{D}_1$ embedded by model $\texttt{emb}_1$ and merges them into database $\mathcal{D}_2$ embedded by model $\texttt{emb}_2$ via function $\mathcal{T}$, in the form of a collection of isometries. Each isometry f^i_{iso} is determined by a local cluster X_i of reference vectors that corresponds to its counterpart Y_i in $\mathcal{D}_2$.

Integration error (β). Consider an arbitrary non-reference vector $\mathbf{x}$ of $\mathcal{D}_1$. We call $\mathcal{T}(\mathbf{x})$ its image *w.r.t.* $\mathcal{T}$ in the space of $\mathcal{D}_2$. Let $o_{\mathbf{x}}$ be the data item that it encodes in $\mathcal{D}_1$. Then the *integration error* $\beta(\mathbf{x})$ of $\mathcal{T}$ is the distance between $\mathcal{T}(\mathbf{x})$ and $\texttt{emb}_2(o_{\mathbf{x}})$, *i.e.,* the imagined "ideal" vector that $\texttt{emb}_2$ would have generated in $\mathcal{D}_2$ if it were asked to embed $o_{\mathbf{x}}$. It should be noted that $\texttt{emb}_2(o_{\mathbf{x}})$ is a hypothetical vector that is not part of $\mathcal{D}_2$, but it resides in the same ambient space.

Note that the integration error β is defined for all vectors of $\mathcal{D}_1$, and reflects the divergence between the integrated vectors and their "ideal" position in the vector space of $\mathcal{D}_2$. As shown in Fig. 5, it is sensitive to how we group references $\mathcal{X}$ of $\mathcal{D}_1$ in step (1) of LA2M.

To reveal their connections, we further delve into the contributing factors of β. In particular, the integration error of $\mathbf{x}$, *i.e.,* $\|\mathcal{T}(\mathbf{x}) - \texttt{emb}_2(o_{\mathbf{x}})\|$, is the combined effect of two sources of errors:

- *Model disagreement*: the inherent difference between $\texttt{emb}_1$ and $\texttt{emb}_2$, *i.e.,* even if both models embed the same dataset, their produced vector databases would not align with each other.
- *Merging error*: the "generalization error" of isometry f^i_{iso} determined by references X_i for transforming and integrating vector $\mathbf{x}$ that are not in X_i.

Intuitively, model disagreement is inherent to embedding models and is hence constant *w.r.t.* any given $\mathcal{D}_1$ and $\mathcal{D}_2$. Computationally, this is reflected by that even the optimal isometry f^i_{iso} cannot exactly align all references in X_i of $\mathcal{D}_1$ to their images in Y_i of $\mathcal{D}_2$ by step (2) of LA2M. Hence, the optimal

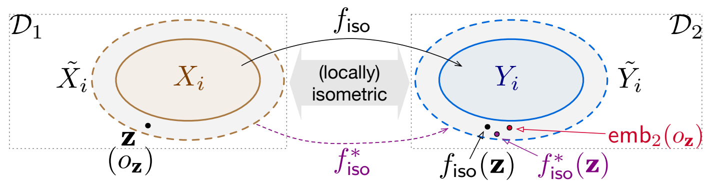

Fig. 6. Errors when using f_{iso} deduced from reference $X_i \rightarrow Y_i$ to map vector $\mathbf{z} \notin X_i$. (1) $f_{\text{iso}}(\mathbf{z})$: mapping of $\mathbf{z}$ by our isometry f_{iso}; (2) $f^*_{\text{iso}}(\mathbf{z})$: $\mathbf{z}$ mapped by the *optimal* isometry f^*_{iso} that agrees with our f_{iso} on X_i. (3) $\texttt{emb}_2(o_{\mathbf{z}})$: the envisaged embedding of $o_{\mathbf{z}}$ by $\texttt{emb}_2$, where $o_{\mathbf{z}}$ is the data item encoded by $\texttt{emb}_1$ in $\mathcal{D}_1$. The distance $\|(1)-(2)\| = \|f_{\text{iso}}(\mathbf{z}) - f^*_{\text{iso}}(\mathbf{z})\|$ corresponds to the merging error (model-independent), while $\|(1)-(3)\| = \|f_{\text{iso}}(\mathbf{z}) - \texttt{emb}_2(o_{\mathbf{z}})\|$ is the integration error (depends on $\texttt{emb}_1$ and $\texttt{emb}_2$).

alignment error of Equation 1 in step (2) of LA2M can be an indicator that helps decide if $\mathcal{D}_1$ and $\mathcal{D}_2$ can be integrated: if there are clusters X_i that cannot be aligned with Y_i with small alignment error, $\mathcal{D}_1$ and $\mathcal{D}_2$ may capture different similarity or irrelevant datasets and thus should not be integrated.

On the other hand, merging error originates from the way we integrate $\mathcal{D}_1$ and $\mathcal{D}_2$ and is what we aim to minimize in LA2M. We next define the notion and deduce an upper bound of the error.

Bounding merging error. As depicted in Fig. 6, the merging error of a non-reference $\mathbf{z}$, denoted by $\delta(\mathbf{z})$, is the distance between $f_{\text{iso}}(\mathbf{z})$ and $f^*_{\text{iso}}(\mathbf{z})$ in the vector space of $\mathcal{D}_2$, *i.e.,* $\|f_{\text{iso}}(\mathbf{z}) - f^*_{\text{iso}}(\mathbf{z})\|$, where

(a) f_{iso} is the isometry determined by LA2M over X_i and Y_i, *i.e.,* the optimal isometry that aligns local references X_i of $\mathcal{D}_1$ to Y_i of $\mathcal{D}_2$ with minimum sum of squared errors (*i.e.,* Equation 1);

(b) f^*_{iso} is the optimal isometry that aligns larger locally isometric regions $\tilde{X}_i$ and $\tilde{Y}_i$ of $\mathcal{D}_1$ and $\mathcal{D}_2$ vector spaces, where $\tilde{X}_i$ consists of reference vectors of X_i and non-reference vectors $\mathbf{z}$ mapped by f_{iso} (in step (3) of LA2M), and $\tilde{Y}_i = Y_i \cup \{\texttt{emb}_2(\mathbf{z}) \mid \mathbf{z} \in \tilde{X}_i \setminus X_i\}$.

Intuitively, f_{iso} is the optimal isometry that LA2M computes in step (2) via references X_i and Y_i, and f^*_{iso} is the optimal isometry that would be determined by mapping the larger reference sets $\tilde{X}_i$ and $\tilde{Y}_i$. Note that $\tilde{Y}_i$ contains hypothetical vectors that $\texttt{emb}_2$ would generate for data items encoded by non-reference vectors integrated by f_{iso} in step (3) of LA2M.

We next state our main result, an upper bound on $\delta(\mathbf{z})$ in terms of (a) the alignment errors of f_{iso} and f^*_{iso} on references X_i and Y_i, and (b) the distance between $\mathbf{z}$ and the centroid $\mathbf{c}_X$ of X_i.

More specifically, the *alignment error* of f_{iso} on X_i and Y_i is the total alignment errors of reference vectors in X_i *w.r.t.* f_{iso}, calculated by $\sum_{\mathbf{x} \in X_i} \|f_{\text{iso}}(\mathbf{x}) - \mathbf{y}\|^2$ (Equation 1), where $\mathbf{x} \in X_i$ corresponds to $\mathbf{y} \in Y_i$, *i.e.,* they encode the same data item; similarly for f^*_{iso}. Denote by α and α^* the alignment errors of f_{iso} and f^*_{iso} on (X_i, Y_i) respectively. For simplicity, in the sequel we write X_i (resp. Y_i) as X (resp. Y). We also treat X and Y as matrices with their vectors as columns.

Theorem 1: *Let $\sigma_{\min}(X)$ be the minimum singular value of X. Assume that X and $\tilde{X}$ (resp. Y and $\tilde{Y}$) have the same centroid. Then:*

$$\delta(\mathbf{z}) \leq \frac{\alpha + \alpha^*}{\sigma_{\min}(X)} \|\mathbf{z} - \mathbf{c}_X + \mathbf{c}_Y\|$$

□

Proof: From Section 3.2 we know that $f_{\text{iso}}(\mathbf{x}) = s * R(\mathbf{x} - \mathbf{c}_X + \mathbf{c}_Y)$ for some rotation matrix R and scaling factor s; similarly $f^*_{\text{iso}}(\mathbf{x}) = s^* R^* (\mathbf{x} - \mathbf{c}_{\tilde{X}} + \mathbf{c}_{\tilde{Y}})$ for some R^* and s^*. Since X and $\tilde{X}$ have overlapping centriod, $\mathbf{c}_X = \mathbf{c}_{\tilde{X}}$; similarly, $\mathbf{c}_Y = \mathbf{c}_{\tilde{Y}}$. Denote by $\hat{X}$ the matrix translated from X (as a matrix) with translation vector $\mathbf{c}_Y - \mathbf{c}_X$.

Let $E = \sqrt{\sum_{\mathbf{x} \in X} \|f_{\text{iso}}(\mathbf{x}) - f^*_{\text{iso}}(\mathbf{x})\|^2}$. Written in matrix form,

$$E = \|(sR - s^*R^*)\hat{X}\|_F \quad \text{(Frobenius norm)}$$

Let singular value decomposition (SVD) of $\hat{X}$ be $\hat{X} = U\Sigma V^\top$ with

$$\Sigma = \text{diag}(\sigma_1, \ldots, \sigma_d), \quad \sigma_1 \geq \cdots \geq \sigma_d \geq 0,$$

and $V = [v_1, \ldots, v_d]$ containing the right singular vectors. Since U is orthogonal, $E = \|(s\,R - s^*R^*)\hat{X}\|_F = \|\Sigma\, V^\top (s\,R - s^*R^*)\|_F$. Taking the square and expanding the Frobenius norm, we have

$$\|\Sigma\, V^\top (s\,R - s^*R^*)\|_F^2 = \sum_{i=1}^{d} \sigma_i^2\, \|v_i^\top (s\,R - s^*R^*)\|^2.$$

Hence, for each right singular direction v_i,

$$\|\,s\,Rv_i - s^*Rv_i\| \leq \frac{\|(s\,R - s^*R^*)X\|_F}{\sigma_i} \leq \frac{E}{\sigma_{\min}(\hat{X})} = \frac{E}{\sigma_{\min}(X)},$$

where $\sigma_{\min}(\hat{X}) = \min_i \sigma_i$ and $\sigma_{\min}(X)$ is the minimum non-zero singular of X (note that $\hat{X}$ and X have the same singular values).

Let $\mathbf{z} \in \mathbb{R}^d$ be a new point in $\tilde{X} \setminus X$. Express its translated version $\hat{\mathbf{z}} = \mathbf{z} + \mathbf{c}_Y - \mathbf{c}_X$ in the basis $\{v_1, \ldots, v_d\}$ as $\hat{\mathbf{z}} = \sum_{i=1}^{d} \alpha_i\, v_i$ with $\alpha_i = v_i^\top \hat{\mathbf{z}}$. Then, its images by f^*_{iso} and f_{iso} in $\mathcal{D}_2$ vector space are

$$f^*_{\text{iso}}(\mathbf{z}) = s^*R^*\hat{\mathbf{z}} = \sum_{i=1}^{d} \alpha_i \left[s^*Rv_i\right],\ f_{\text{iso}}(\mathbf{z}) = s\,R\hat{\mathbf{z}} = \sum_{i=1}^{d} \alpha_i \left[s\,Rv_i\right].$$

Thus, their difference is given by

$$s^*R^*\hat{\mathbf{z}} - s\,R\hat{\mathbf{z}} = \sum_{i=1}^{d} \alpha_i \left[s^*Rv_i - s\,Rv_i\right].$$

Taking norms and applying the triangle inequality,

$$\|s^*R^*\hat{\mathbf{z}} - s\,R\hat{\mathbf{z}}\| \leq \sum_{i=1}^{d} |\alpha_i|\, \|s^*Rv_i - s\,Rv_i\| \leq \frac{E}{\sigma_{\min}(X)} \sum_{i=1}^{d} |\alpha_i|.$$

By an application of the Cauchy–Schwarz inequality, we know $\sum_{i=1}^{d} |\alpha_i| \leq \sqrt{d}\, \|\hat{\mathbf{z}}\|$. Thus, we may write a succinct bound:

$$\|f^*_{\text{iso}}(\mathbf{z}) - f_{\text{iso}}(\mathbf{z})\| = \|s^*R^*\hat{\mathbf{z}} - s\,R\hat{\mathbf{z}}\| \leq \frac{E}{\sigma_{\min}(X)}\, \|\mathbf{z} - \mathbf{c}_X + \mathbf{c}_Y\|.$$

By the triangle inequality of Frobenius norm, we further have

$$E \leq \|sR\hat{X} - Y\|_F + \|s^*R^*\hat{X} - Y\|_F = \alpha + \alpha^*.$$

Putting together, we have $\delta(\mathbf{z}) \leq \dfrac{\alpha + \alpha^*}{\sigma_{\min}(X)} \|\mathbf{z} - \mathbf{c}_X + \mathbf{c}_Y\|$. □

Remark. To simplify the discussion, the upper bound assumes that X is full rank, *i.e.*, its singular values are all non-zero. This suffices to offer insights to inform the design specification of LA2M. As will be shown in Section 6, our design of LA2M works with both full rank and rank-deficient X.

Drawing insight from bound. Theorem 1 tells us the following.

(**I1**) Minimizing the alignment error α of f_{iso} over references X and Y helps reduce the merging error of non-reference vectors. This also justifies the objective function of Equation 1 and its application in step (2) of LA2M. As shown in Fig. 5 (blue line), the smaller X is, the lower α could be.

(**I2**) However, simply reducing X does not work. When X is too small, the minimum singular value $\sigma_{\min}(X)$ of its matrix form would be extremely small or even 0, which means $\delta(\mathbf{z})$ can be unbounded or infinitely large, despite that α is small. This is consistent with Fig. 5 where the overall integration error β is the worst when n approaches its maximum 1000, despite that α reaches its minimum.

(**I3**) This further tells us that poor conditioning of X (*i.e.,* small $\sigma_{\min}(X)$) leads to significantly amplified merging error. Moreover, directions of X with low variance (*i.e.,* small σ_i) are particularly susceptible to rotation ambiguity that elevates $\delta(z)$ hyperbolically.

(**I4**) The merging error $\delta(\mathbf{z})$ is also related to the position of $\mathbf{z}$ relative to the centroids of X and Y. The translation of X into $\hat{X}$ in the proof of Theorem 1 also implies that, if we center X and Y to their centriods first, then we would not need translation in f_{iso} and f^*_{iso}. Moreover, the upper bound of $\delta(\mathbf{z})$ would be then simplified to $\delta(\mathbf{z}) \leq \dfrac{\alpha + \alpha^*}{\sigma_{\min}(X)} \|\mathbf{z}\|$. As such, the closer $\mathbf{z}$ is to the translated centroid of X (*i.e.,* the origin) the smaller the merging error. In fact, this is exactly how we implemented step (2) of LA2M. This also justifies why we pick $f^{j^*}_{\text{iso}}$ that has the closest centroid when we merge a new non-reference vector $\mathbf{z}$ in step (3) of LA2M.

6 Informing the Framework Design

We next present the design choices of LA2M based on the insights offered by the bound of Section 5.

6.1 Breaking down the Global Isometry

We first consider step (1) of LA2M, by studying how we break down $\mathcal{X}$ into local groups to determine the local isometries of step (2). We aim to minimize the merging error in step (3) of LA2M.

Locality Vs. **dimensionality**. Consider a non-reference vector $\mathbf{z}$ of $\mathcal{D}_1$. According to (**I4**) of Section 5, to merge $\mathbf{z}$ into $\mathcal{D}_2$ with a minimum merging error, we shall use an isometry based on references X with a centroid as close to $\mathbf{z}$ as possible. By (**I1**), X should be as small as possible to minimize alignment error α. Further according to **I2** and **I3**, this should not make X rank-deficient, *i.e.,* the size $|X|$ should be at least d, where d is the dimensionality of the embeddings. These suggest that the ideal X for merging $\mathbf{z}$ would consists of at least d reference vectors of $\mathcal{D}_1$ that are closest to $\mathbf{z}$.

However, real-life embedding vectors often have high dimensionality. For instance, `OpenAI-Ada` embeddings are of 1536 dimension and that of `Mistral` is 1024. Consequently, this could necessitate large X and may inadvertently result in significant alignment errors. These, in turn, could lead to suboptimal merging errors. In addition, even if we have a large full rank X, its size might lead to nonzero but rather small singular values, exacerbating merging errors further.

Indeed, (**I3**) naturally motivates us to consider reducing the dimensionality of vectors via PCA (Principal Component Analysis [2]) to avoid poor conditioning of X, in particular eliminating directions with low variance (small singular values σ_i). This would give us an isometry f_{iso} based on X that is insusceptible to embedding noise.

Thus, determining the size of X reduces to finding the right dimensionality for the PCA-reduced reference vectors. Recall that to allow step (2) of LA2M to determine a local isometry from X to Y that can accurately bind vectors of $\mathcal{D}_1$ that are close to X, one has to assure X contains vectors that are able to determine inherent (*i.e.,* minimum) dimensionality of dissimilarity among data items encoded by X as well as those would be transformed by the local isometry determined by X. While we cannot exactly know the inherent dimensions, we however could approximate it with *intrinsic dimension* estimation over the reference vectors using Maximum Likelihood Estimation (MLE) [51].

The intrinsic dimension characterizes the minimal number of dimensions necessary to effectively represent manifold structures of data without significant information loss. The MLE-based intrinsic

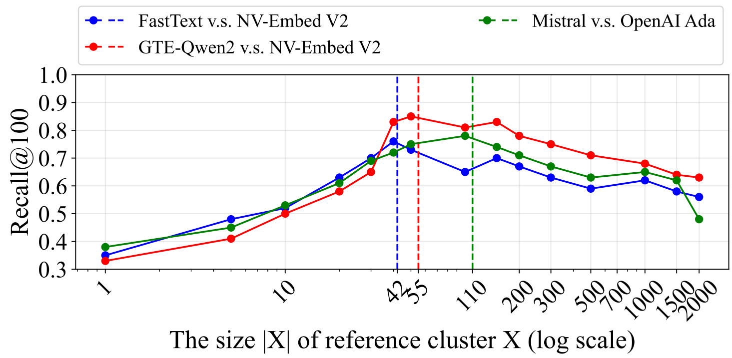

Fig. 7. Integration accuracy (recall@100) Vs **size of local reference clusters** X **over** SciFact**:** the vertical dashed lines mark the intrinsic dimensions. Benchmark results verify that the accuracy (recall@100) peaks when $|X|$ is close to the intrinsic dimension in all three integration cases.

dimension estimator determines this optimal intrinsic dimensionality of a given data point (vector) by analyzing its local neighborhoods. For each data point, the local intrinsic dimension is computed by examining the distances to its k-nearest neighbors. The intrinsic dimensionality $\hat{d}$ can be obtained via the MLE estimator[51]: $\hat{d} = \left[\frac{1}{N(K-1)} \sum_{i=1}^{N} \sum_{j=1}^{K-1} \log \frac{T_K(x_i)}{T_j(x_i)}\right]^{-1}$, where $T_j(x_i)$ denotes the distance from data point x_i to its j-th nearest neighbor, and N is the total number of data points of concern.

Mapping to our case, we set N to 10 and K to 20 for each dataset, and use $\hat{d}$ of the reference vector $\mathbf{x}$ closest to $\mathbf{z}$ to determine the size of X. In cases where we only know reference vectors, we use the average intrinsic dimensionality of randomly sampled reference vectors as a database-wide configuration of the size of X. Using benchmarks, results in Fig. 7 confirm that intrinsic dimension does serve as a quite accurate estimation of the best size of X for determining local isometries.

Computing with rank-deficient references. While we can use intrinsic dimension of reference vectors to decide the size of reference cluster X, it however requires us to reduce the dimensions of embedding vectors. To avoid directly altering vector databases for merging, we improve upon A2M with PCA in step (2) of LA2M, such that it works with small X containing high-dimensional embedding vectors, *i.e.,* rank-deficient reference clusters X.

Consider a pair of corresponding reference clusters $(X, Y) \in \mathbb{R}^{m\times d}$. We first perform SVD on X and Y, yielding $\mathrm{SVD}(X) = U_X \Sigma_X V_X^\top$ and $\mathrm{SVD}(Y) = U_Y \Sigma_Y V_Y^\top$. We then project X and Y into their respective $\hat{d}$-dimensional subspaces: $\hat{X} = XV_X^{\hat{d}}, \hat{Y} = YV_Y^{\hat{d}}$, where $\hat{d}$ is the size of X (*i.e.,* intrinsic dimensionality determined above), $V_X^{\hat{d}}, V_Y^{\hat{d}} \in \mathbb{R}^{d\times\hat{d}}$ are the top $\hat{d}$ right singular vectors of X and Y, respectively. We rewrite the objective function in Equation 1 as:

$$\underset{s,R,t}{\arg\min} \left\| sR(\hat{X} + t) - \hat{Y} \right\|_F^2. \tag{2}$$

Let $\hat{f}_{\mathrm{iso}}$ be the optimal isometry determined by $\hat{X}$ and $\hat{Y}$. For a new vector $\mathbf{z} \in \mathbb{R}^{1\times d}$, we first reduce its dimensionality using $V_X^{d'}$, yielding $\hat{\mathbf{z}} = \mathbf{z}V_X^{d'}$. We then apply $\hat{f}_{\mathrm{iso}}$ to $\hat{\mathbf{z}}$ and finally reconstruct the transformed vector by multiplying with $V_Y^{d'\top}$: $\hat{f}_{\mathrm{iso}}(\hat{\mathbf{z}})V_Y^{d'\top}$.

Online integration. Recall from Section 4.2 that steps (1) and (2) of LA2M work on reference vectors, independent of non-reference vectors of $\mathcal{D}_1$ that need to be merged into $\mathcal{D}_2$. In cases where there are constantly new vectors added to $\mathcal{D}_1$, LA2M separates step (3) from the first two steps, to reduce the time of merging these new vectors.

Table 2. Statistics of benchmarks

Dataset	#Train Pairs	#Test Queries	#Test Corpus Size
SciFact [80]	920	300	5,183
NFCorpus [9]	110,575	323	3,633
ArguAna [79]	n/a	1,406	8,674
SciDocs [18]	n/a	1,000	25,657
FiQA-2018 [59]	14,166	648	57,638

Specifically, as offline preprocessing, LA2M carries out steps (1) and (2) and materializes local isometries $f^1_{\text{iso}}, \ldots, f^n_{\text{iso}}$ as a vector database itself, where each vector is the centroid of a reference cluster X_i in $\mathcal{D}_1$, of which the encoded datum is the isometry f^i_{iso}.

LA2M merges non-reference vectors of $\mathcal{D}_1$ as an online process. Once a non-reference vector $\mathbf{x}$ is available or newly added to $\mathcal{D}_1$, LA2M finds the closest centroid $\mathbf{c}_j$ via a top-1 vector search over the centroids, and retrieves the corresponding isometry f^i_{iso} to merge $\mathbf{x}$.

To minimize merging error of new vectors of $\mathcal{D}_1$, we compute and materialize an isometry for each reference vector $\mathbf{x}$ in $\mathcal{X}$ of $\mathcal{D}_1$: we first find $\hat{d}$ nearest neighboring references of $\mathbf{x}$ in $\mathcal{X}$, forming a cluster X with $\mathbf{x}$ approximately as its centriod. We then compute the isometry f^X_{iso} determined by X. Hence, we have $n = |\mathcal{X}|$ number of isometries, each indexed by a reference vector $\mathbf{x} \in \mathcal{X}$ as its approximate centroid. In case we want to limit the number of materialized isometries, we iteratively merge pair of closest isometries:we take the union of their reference clusters and replace the pair of isometries with the one determined by their combined reference vectors.

6.2 Making Embeddings Mergeable

We next discuss the availability of reference vectors, *i.e.,* pairs of vectors $(\mathbf{x}, \mathbf{y})$ such that $\mathbf{x} \in \mathcal{D}_1$ and $\mathbf{y} \in \mathcal{D}_2$ encode the same data item. Naturally, known correspondence in embedding backfilling, entity linkage and domain knowledge are viable sources of reference vectors across datasets. However, in case they are not given as a priori, one may still benefit from LA2M by actively generating reference vectors with limited access to the embedding models.

More specifically, let $\mathcal{D}$ be a vector database that encode data items of $\mathcal{O}$. Assume that we need to integrate $\mathcal{D}$ into another (possibly unknown) vector database that offers access to their embedding model emb. We consider the task of selecting up to $\gamma * |\mathcal{D}|$ data items of $\mathcal{O}$ (vectors of $\mathcal{D}$) as reference items with some $\gamma \in (0, 1]$, to form reference vector pairs by requesting their encoding of emb.

By (**I4**) of Section 5, we know that ideally these $\gamma|\mathcal{D}|$ reference items shall cover all items of $\mathcal{O}$ with close proximity in $\mathcal{D}$. In light of this, we propose a reference sampling method that operates on $\mathcal{D}$. It is parameterized by L, the number of local isometries we want to generate, and works as follows.

(1) It first forms L clusters of $\mathcal{D}$ via k-means clustering.
(2) For each cluster C_i, it uniformly samples $\gamma|C_i|$ vectors as references. In total this samples a set $\mathcal{X}$ of $\gamma|\mathcal{D}|$ vectors from $\mathcal{D}$.
(3) It queries emb with the data items of $\mathcal{O}$ encoded by $\mathcal{X}$, yielding a set $\mathcal{Y}$ of corresponding vectors in the target vector database. The pair $(\mathcal{X}, \mathcal{Y})$ forms references that enable us to integrate $\mathcal{D}$ into the targeted vector database with LA2M.

Remark. For cases where one wants to avoid revealing any item in $\mathcal{O}$, one may use synthesized objects as the references. It however works best if $\mathcal{D}$ is generated by , *e.g.,* a diffusion embedding model which allows us to synthesize reference items that can cover $\mathcal{O}$ well.